

Job Market Analysis Report

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Date: 24/05/2026

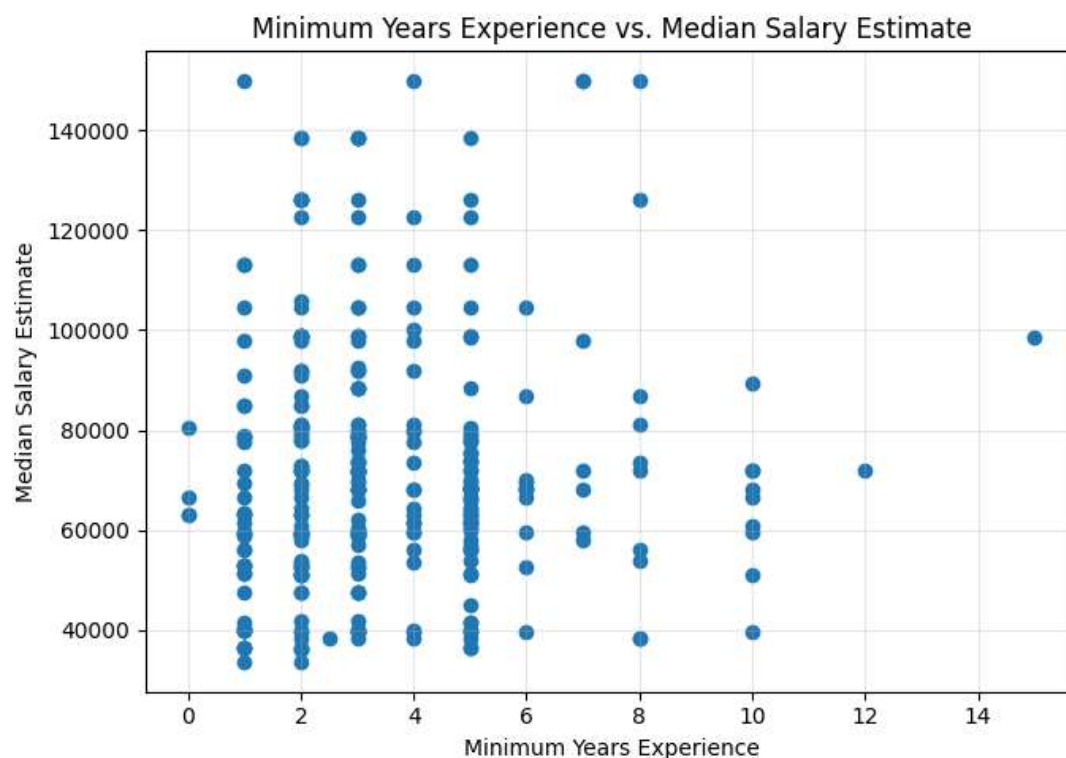
Introduction

This project analyzes a sample of 400 Data Analyst job postings. The data includes salary estimates, extracted minimum years of experience, and categorized programming language requirements. Before plotting, the “Minimum Years Experience” field was cleaned so non-numeric values such as “NA” were treated as missing, and “Data Languages” was standardized into four categories: R, Python, Both, and Neither.

Plot 1 Interpretation: Experience vs. Salary

The scatter plot shows the relationship between minimum years of experience and median salary estimate. Most postings are concentrated between 1 and 5 years of experience, with salaries spread widely across that range. There is no strong visible linear relationship in this sample, although some higher salaries appear among postings requiring several years of experience.

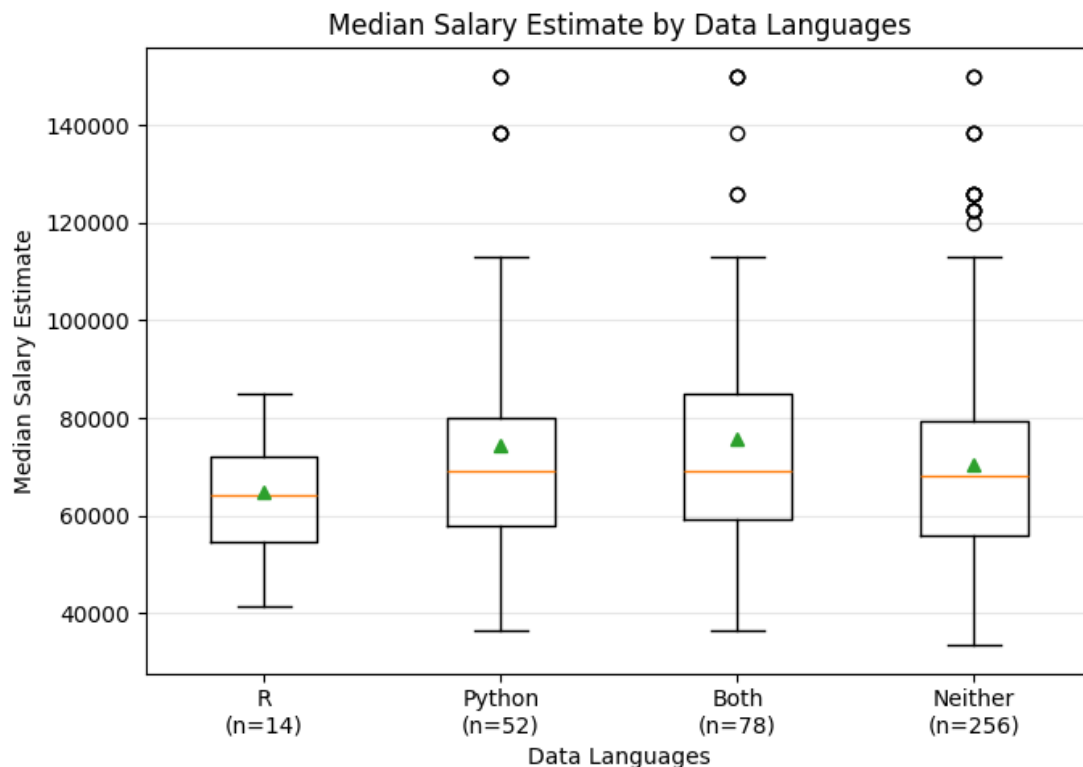
A limitation is that only 310 of the 400 postings had usable numeric experience values, so the plot excludes postings where experience could not be extracted cleanly.



Plot 2 Interpretation: Salary by Data Languages

The box plot compares median salary estimates across postings requiring R, Python, both languages, or neither. Python, Both, and Neither have fairly similar salary distributions, while R has a much smaller sample size. Some high-salary outliers appear in the Python, Both, and Neither groups.

Because the category sizes are uneven, especially R with only 14 postings, these differences should be interpreted cautiously. The sample is relatively small and may not represent the broader Data Analyst job market.



The personal reflection :

The first section of the assignment, which was about using the LLM API to find the minimum years of experience and to identify what programming languages mentioned in the Job Description column, was challenging, mainly when I had to create the prompt on my own, specifically the formula (where to place the “ “ and () and , .

The second part, when using ChatGPT to generate the graphs was easy, mainly because the prompts are available and accessible. It facilitates the work tremendously.